TASK 12 – RE-EVALUATE AND ADAPT THE DESIGN

1. Aim

To modify the existing milk payment calculator to support two-decimal fat and SNF readings, seasonal rate changes and a ten-day supply bonus.

2. Introduction

The original calculator used a fixed base rate of ₹30.00 per litre, a fat step value of ₹0.40 and an SNF step value of ₹0.30.

The new requirements introduce:

- Fat and SNF readings with two decimal places.
- A seasonal rate card from April to June.
- A bonus for farmers supplying more than 500 litres in ten days.

The existing calculator must be modified to support these changes.

3. A. Impact Analysis (2 Marks)

Changes in the requirements

Feature	Original requirement	New requirement
Fat and SNF	One decimal step values	Readings with two decimal places
Base rate	₹30.00 throughout the year	₹32.00 from April to June
Fat step value	₹0.40	₹0.45 from April to June
SNF step value	₹0.30	₹0.30 throughout the year
Bonus	No bonus	₹0.50 per litre if supply exceeds 500 litres in ten days

Constants affected

The following constants must be updated or selected according to the month:

- Base rate
- Fat step value
- SNF step value
- Bonus rate
- Bonus threshold

Functions affected

The rate calculation function must accept the month and select the correct rate card.

A separate bonus function is required to check whether the farmer's ten-day supply is greater than 500 litres.

Effect of two-decimal readings

The calculator must accept readings such as 4.25% fat and 8.63% SNF.

The rate adjustment is based on 0.1% steps. Therefore, the number of steps is calculated and rounded to the nearest whole step.

For this implementation, ROUND_HALF_UP is used when a reading falls halfway between two 0.1% steps.

Example:

- Fat: 4.25%
- Standard fat: 3.50%
- Difference: 0.75%
- Fat steps: 7.5, rounded to 8.

For SNF:

- SNF: 8.63%
- Standard SNF: 8.50%
- Difference: 0.13%
- SNF steps: 1.3, rounded to 1.

4. B. Evaluation of the Original Design (2 Marks)

The original design already uses Decimal, reusable functions and a separate rate card.

These features remain useful because they support accurate financial calculations and make the program easier to maintain.

However, the original fixed constants are no longer sufficient because the base rate and fat step value change during April, May and June.

The original rate calculation also needs to support two-decimal readings and a defined step-rounding rule.

The original design can therefore be retained with modifications.

Worked example

For a reading of 4.25% fat and 8.63% SNF:

Fat steps = 8

SNF steps = 1

April–June:

Rate = 32.00 + (8 × 0.45) + (1 × 0.30)

Rate = ₹35.90 per litre.

Other months:

Rate = 30.00 + (8 × 0.40) + (1 × 0.30)

Rate = ₹33.50 per litre.

This shows that the seasonal rate selection changes the payment calculation.

5. C. Adaptation or Redesign (2 Marks)

The existing design is modified using the following functions:

1. get_rate_card() – selects the seasonal rate card.
2. calculate_rate() – calculates the price per litre.
3. calculate_payment() – calculates the payment for a